

REPORT 3

This report discusses the following topics:

Analog vs Digital signals

Ac vs Dc (And PWM)

Electronic components

ANALOG VS DIGITAL SIGNALS:

DIGITAL SIGNALS:

These signals show a variation of discrete values of voltage(and current). That is the electric component of the signal is in bits (usually 0s and 1s). This type of signal is used by electronic components as they process signals in binary and hence need a digital signal

ANALOG SIGNALS:

These signals are continuous signals which have a quantity which varies with time(often voltage or current). Analog signals are used in places where a continuous value is required. Examples would be: telephone, mobile phones(communication) etc.

ENCODING AND DECODING:

In a communication system, when a signal is transmitted through a medium, it encounters a lot of other signals which are not useful to us. Therefore, the signal is encoded with a particular code so that it can be retrieved at the receiver side by decoding the received signal.

encoding can be of a few types:

Analog to analog: where the input signal is analog as well as the encoded signal is analog. This is used in telephonic communication

Digital to analog: In this, the digital signal is “ modulated” by an analog signal(Frequency modulation amplitude modulation etc) this effectively encodes the signal which can be decoded later to retrieve the original signal. Mostly used in cases where digital data has to be transmitted through a wireless or wired connection.example: modem.

Analog to Digital: In this case, the analog data is encoded by the digital signal. That is the continuous shape of the wave is modelled into a square wave of 0s and 1s(bits).

An example would be to convert the received data (data from a sound sensor etc) into a digital form. Process of converting analog data to digital data is called digitisations

AC VS DC AND PWM:

Ac: alternating current, Dc: Direct current, PWM: Pulse width modulation

AC is a type of current where the voltage of the current varies with time. The variation might be sinusoidal, or any other curve, it can even be discrete. It is AC as long as the value representing the current alternates or oscillates

DC is the type of current which does not vary with time. That is, all the characteristics of the current are constant with respect to time.

PWM or pulse width modulation is a method of encoding analog data with a digital signal.

The digital signal represents the analog signal by varying the time for high or low voltage for calculated time intervals which if decoded would give back an analog signal particular to the given digital signal as the time it is in high voltage represents the frequency and amplitude as well as the curve of the analog signal.

This is used to remove many of the disadvantages analog signal has over digital signal which include higher perceptibility to interference, heating up of circuit due to power dissipation etc

ELECTRONIC COMPONENTS:

The electronic components in the rover would be microcontrollers(as the control system), the processor for machine learning application and various sensors.

MICROCONTROLLERS:

Microcontrollers are microprocessors or micro computers which unlike the microprocessors have a specific task they are meant to do.

Microcontrollers also come with integrated RAM and ROM as well as other peripherals (programmable input and output peripherals) and are usually programmable or can be preprogrammed.

Microcontrollers are used because the integrated peripherals they come with allow them able to act as a very small computer in itself which is able to carry out instructions

specific to its specifications. Microcontrollers come in different types, one of the major types they are divided in is based on the size of the instruction set they can work with or how much memory they can work with. For instance, an 8 bit microcontroller can work with 8 bit wide instruction sets(at once) or 8 bit wide memory locations. Similarly, the 32 bit can work upto 32 bit wide instruction sets. The onboard microcontroller also acts as the control system which helps the driver not only precisely maneuver the rover by processing the instructions sent and carrying them out, but also perform other tasks (as in using robotic arm, etc). The microcontroller is also used for data acquisition for the data from the sensors.

SENSORS:

The rover requires a lot of sensors, a few of them would be accelerometers, magnetometers, gyroscopes etc for finding out orientation, cameras, environmental sensors for finding out environmental conditions such as wind speed, temperature etc

These sensors are all connected to the microcontroller and the autonomous system. The microcontroller acts as the control system and leads to a lot of the onboard processing.

The accelerometer and the magnetometer help finding the orientation of the rover. This helps in rover communication and data collection. These achieve their tasks in different ways. The accelerometer measures the acceleration on the body and hence finds its position, velocity and orientation(gravity also accelerates) while the magnetometer uses the magnetic fields to find out the orientation of the rover. This data is sent to the base station or the autonomous system for analysis and detection.

The cameras aboard the rover act as eyes which help it not only in taking pictures but also with finding out depth. The hazard cams which are close to the surface scan for obstacles which would be hazardous to the rover should it try and traverse it. The cameras only take the pictures/videos which is sent to the base station or the autonomous system(onboard) for detection and analysis

The environmental sensors which include wind speed sensors and pressure and temperature sensors are digital tachometers, barometers, thermometers etc which send data to the base station and the autonomous system for analysis.

AUTONOMOUS SYSTEM:

The autonomous system aboard the Mars rover manipulator is built on the Nvidia Jetson Tx1 which is a GPU with an integrated board. That is, just like a microcontroller, it comes with its own peripherals, RAM and onboard internal storage. The Jetson Tx1 is used for many machine learning application tasks as it is portable and the processing power it possesses is enough for most ML AI tasks .

The artificial intelligence onboard the rover has a big task of avoiding hazards in its path.

It uses the photos and videos by the mast cam to map out a 3D image of its surroundings and then by using image processing software, the artificial intelligence in board maps out the safest shortest path(the path avoiding all obstacles). It also uses the real time feed from the haz cams along with its image processing capabilities to avoid obstacles which may damage the rover in case traversal is attempted.

The artificial intelligence onboard is also used for sensor data analysis which include analysis of orientation (should the data be required by the base station) and all environmental sensors. A part of the soil analysis is carried onsite.

ANTENNAS:

There are two types of antennas used for MRMs rover. Omnidirectional and unidirectional.

The unidirectional antenna is used at the base station and the omnidirectional antenna is used by the rover.